MULTIPLICATION BY AN ELEMENT OUTSIDE THE MINIMAL PRIMES IS SURJECTIVE ON THE TOP LOCAL COHOMOLOGY OF EVERY FINITE MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The vanishing of local cohomology above the dimension of the support, and the bound cd(A, I) \leq dim(A) for a finitely generated ideal. Author lemma lemma-cd-dimension, whose statement is reproduced verbatim as a context span of this card (source lines 758-764). Supplies both the vanishing on the quotient by f and the bound that produces the second surjection.
- The exact sequence relating the local cohomology of a module, of its image under multiplication by an element and of the quotient. Used inside the counted proof at source lines 3640-3645. Turns the vanishing on the quotient into the required surjectivity.

Lemma 0.1. *Let A be a Noetherian local ring of dimension d. Let $f \in A$ be an element which is not contained in any minimal prime of dimension d. Then $f : H^d_{V(I)}(M) \to H^d_{V(I)}(M)$ is surjective for any finite A-module M and any ideal $I \subset A$.*

Lemma 0.2. *Let $I \subset A$ be a finitely generated ideal of a ring A. If M is a finite A-module, then $H^i_{V(I)}(M) = 0$ for $i > \dim(Supp(M))$. In particular, we have $cd(A, I) \leq \dim(A)$.*

Proof. The support of M/fM has dimension $< d$ by our assumption on f. Thus $H^d_{V(I)}(M/fM) = 0$ by Lemma 0.2. Thus $H^d_{V(I)}(fM) \to H^d_{V(I)}(M)$ is surjective. Since by Lemma 0.2 we know $\mathrm{cd}(A, I) \leq d$ we also see that the surjection $M \to fM$, $x \mapsto fx$ induces a surjection $H^d_{V(I)}(M) \to H^d_{V(I)}(fM)$. □

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